

SARO Rule-Pack Validation Report

****Generated**** by `scripts/generate\_validation\_report.py` from the confusion-matrix artifact (STORY-378). Every rate below is measured, not asserted ? regenerating this report cannot state a number the harness did not produce.

****Generated:**** 2026-07-22

****Corpus tier measured:**** T1 (synthetic-deterministic)

****Corpus records:**** 11

****Completion bar:**** SIGNED

What this measures

For each rule-pack, the rate at which its rules fire when they should (recall) and fire only when they should (precision), measured against a labeled corpus. These are ****analytical measurements of SARO's own rule behaviour**** - they are not a determination about any customer system, and SARO does not issue verdicts.

Results

Rule pack	Precision	Recall	F1	TP	FP	FN	TN
RP-OBS-COMPLETE@1.0.0	1.0	1.0	1.0	7	0	0	37
RP-TOOL-SCOPE@1.0.0	1.0	1.0	1.0	3	0	0	30

Per-rule detail

****RP-OBS-COMPLETE@1.0.0****

Rule	Precision	Recall	TP	FP	FN	TN
OBS-ERROR-INVOCATION-1	1.0	1.0	2	0	0	9
OBS-REQUIRED-FIELDS-1	1.0	1.0	1	0	0	10
OBS-TOKEN-COUNTS-1	1.0	1.0	2	0	0	9
OBS-TRUNCATED-OUTPUT-1	1.0	1.0	2	0	0	9

****RP-TOOL-SCOPE@1.0.0****

Rule	Precision	Recall	TP	FP	FN	TN
TOOL-POLICY-ABSENT-1	1.0	1.0	1	0	0	10
TOOL-SCOPE-OFFERED-1	1.0	1.0	1	0	0	10
TOOL-SCOPE-VIOLATION-1	1.0	1.0	1	0	0	10

Limitations - read this before relying on the numbers above

Understatement over overstatement. These results are real but narrow:

****Only tier T1 was measured.**** Tiers not yet covered: T2 (synthetic-adversarial), T3 (offline labeled), T4 (pilot labeled). A perfect score on a synthetic tier does not demonstrate behaviour on adversarial or real-world data.

****The corpus is synthetic.**** It is deterministic and labeled by construction, which makes it reproducible - but it is not real customer traffic. Real-world (T3) and pilot (T4) data are how these rates are validated against reality, and that data is not yet in place.

****2 rule-pack(s)**** were measured: RP-OBS-COMPLETE@1.0.0, RP-TOOL-SCOPE@1.0.0. Other packs are not covered by this report.

****The completion bar is signed, but coverage is tier-limited.**** The bar (Profile A_recall_weighted) is met on the tier measured here (T1) - but a pass on T1 is ****not full validation****. The bar is signed for T1-T3 and its numeric thresholds are provisional until validated against real (T3/T4) data; T4 thresholds are set jointly with the pilot partner and are not backfilled here.

****Adapters not yet validated on real data:**** the observation adapters (Azure OpenAI, Vertex AI) are exercised by the conformance suite but their FP/FN rates on real provider logs are not yet measured.

How these numbers were produced

The rule engine is deterministic and makes no external model calls (INV-1). The labeled corpus is built by ``scripts/build_validation_corpus.py`` and verified byte-identical in CI, so the confusion matrix is reproducible run to run. This report is regenerated from that artifact; it contains no hand-entered figures.

This is a report of SARO's own rule-pack measurements for a technical reviewer. It carries no regulatory weight, is not a determination under any framework, and includes no customer results. Human review by qualified personnel is required before any regulatory submission.